

Maximal Semantic Clique Enumeration in Vector-Attributed Graphs

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ABSTRACT

Vector-attributed graphs encode two complementary forms of group cohesion: edges record structural relations, while vertex vectors describe semantic compatibility. Maximal clique enumeration captures the former but provides no criterion for the latter. We study maximal semantic clique (MSC) enumeration, which retains complete connectivity and requires the average pairwise vector similarity to reach a threshold. Unlike an every-pair threshold, this aggregate criterion tolerates a small number of atypical pairs when the group remains collectively coherent. It is also non-monotone over clique containment, so structural pivot pruning and early rejection of infeasible partial cliques are not directly applicable. We propose two efficient algorithms with different advantageous regimes under this constraint. STRSUB restores structural pivoting by separating structural MCE from bounded semantic subset enumeration. MONOSEMMCE uses a feasible-parent theorem: deleting a deterministic minimum-contribution vertex from any feasible clique preserves feasibility. The induced canonical forest generates each feasible state once without a visited table or materialized size layers. Across three naturally attributed and five controlled vector-attributed graphs, the two specialized methods jointly complete all 24 workloads, whereas the SEMBK baseline reaches the 1,200-second limit on 15. At q_{99} , MONOSEMMCE is $2.92\times-6.24\times$ faster than STRSUB. On two real graphs, pairwise-threshold MCE retains no reference MSC of size at least ten on GitHub-Social and only 0.32% on ogbn-arxiv-25K, demonstrating the model-level fragmentation avoided by the average constraint.

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The source code, data, and/or other artifacts have been made available at <https://github.com/CKTtari/maximal-semantic-clique-enumeration>.

1 INTRODUCTION

Graphs with vector-valued attributes are increasingly common in data management: social networks associate users with profile features, knowledge graphs attach text embeddings to entities, and citation graphs represent papers by content vectors. Prior work consequently combines dense structure with attributes in correlated-pattern mining [30] and similarity-constrained biclique enumeration [37]. These tasks expose a basic modeling gap: adjacency establishes interaction, whereas vector similarity establishes semantic compatibility, and neither signal alone identifies groups satisfying both requirements.

The maximal clique is the strongest structural model of cohesion: every pair of vertices is connected. In a vector-attributed graph, this structural contract provides a natural candidate group, and the attributes then determine whether its members are collectively compatible. Partitioning and query-oriented attributed-community methods answer different tasks because they need not preserve pairwise adjacency or enumerate every qualifying group. Our objective is therefore to retain the clique contract while adding a group-level semantic criterion to each reported pattern.

The MSC Problem. We study the problem of enumerating *maximal semantic cliques* (MSCs) in a vector-attributed graph. Each vertex carries a real-valued attribute vector; pairwise semantic similarity is measured by cosine similarity. Given a threshold τ , a clique is a τ -semantic clique if its average pairwise similarity is at least τ . An MSC is an inclusion-wise maximal τ -semantic clique: no strict superset is also a τ -semantic clique. The task is to enumerate *all* MSCs of the input graph.

The average constraint is deliberate. An every-pair threshold treats one weak pair as a veto and can fragment a structurally complete group even when the remaining pairs provide strong collective evidence. The average instead permits a limited number of atypical pairs while exposing the total semantic requirement as one auditable quantity. This robustness changes the algorithmic problem: adding a vertex may raise or lower the average, and an infeasible clique may have a feasible strict superset. Feasibility is therefore neither downward nor upward closed over clique containment.

For example, a co-authorship clique spanning information theory and natural-language processing can include two cross-field authors whose vector similarity falls below τ , although the remaining collaboration relations support a coherent research team. The average threshold retains this group-level evidence without requiring every relation to have identical semantic strength.

Why Classical MCE Does Not Suffice. Classical maximal clique enumeration (MCE) reports only cliques that are maximal under adjacency. An MSC, however, may be a proper subset of a structural

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maximal clique: adding another adjacent vertex preserves the clique but can lower its average similarity below τ . Consequently, filtering the MCE output omits valid MSCs that are never reported as structural maxima. The same distinction invalidates a direct transfer of the Bron–Kerbosch (BK) pivot rule. The pivot suppresses branches whose structural answers can be extended by the pivot, whereas that extension need not remain semantically feasible. Moreover, an infeasible partial clique cannot be discarded because subsequent additions may restore its average. Thus the average constraint changes the search space, not merely the cost of validating a structural answer: a direct adaptation must explore pivot-suppressed subcliques and postpone threshold-based rejection until semantic feasibility can be established safely.

These observations also imply that no single search organization dominates all thresholds. At permissive thresholds, structural decomposition amortizes semantic work over relatively few host cliques. At selective thresholds, enumerating every structural maximal clique wastes work on semantically barren regions, while a feasibility-driven search can contract rapidly. Our algorithms are designed around this threshold-dependent change in the search space.

We make the following contributions.

Problem formulation. We formalize exact MSC enumeration under inclusion-wise maximal average similarity, prove the associated size decision problem NP-complete, and characterize why neither structural MCE followed by filtering nor direct BK pivoting is complete for this output family.

Enumeration algorithms. We develop STRSUB, which separates structural MCE from bounded semantic subset enumeration, and MONOSEMCE, which turns the feasible minimum-contribution deletion property into a canonical reverse-search forest. They target permissive and selective thresholds, respectively, while returning the same complete MSC output set.

Large-scale evaluation. We evaluate exactness, runtime, memory, scalability, search-space reduction, and output behavior on three naturally attributed and five controlled vector-attributed graphs under a fixed similarity-quantile protocol. The experiments identify the operating regions of both searches, isolate their mechanisms, and compare the MSC definition with pairwise-threshold and relaxed structural pattern families.

2 PRELIMINARIES

2.1 Notations and Problem Definition

We study a *vector-attributed graph* $\mathcal{G} = (V, E, \mathbf{X})$, where V is the vertex set, $E \subseteq \binom{V}{2}$ is the edge set, and $\mathbf{X} = \{\mathbf{x}_v \in \mathbb{R}^D \mid v \in V\}$ assigns a D -dimensional attribute vector to each vertex.

DEFINITION 1 (SIMILARITY FUNCTION). A similarity function $\text{sim} : \mathbb{R}^D \times \mathbb{R}^D \rightarrow [-1, 1]$ is a symmetric, bounded function measuring the semantic closeness of two attribute vectors. We assume all attribute vectors are ℓ_2 -normalized, so that sim coincides with cosine similarity and its range is naturally $[-1, 1]$.

Throughout this paper we work with normalized vectors and precompute $\text{sim}(u, v)$ for all $(u, v) \in E$. The threshold $\tau \in [-1, 1]$ is drawn from the same range.

DEFINITION 2 (CLIQUE). A set $C \subseteq V$ is a clique in $\mathcal{G}$ if every pair of distinct vertices in C is connected by an edge, i.e., $\{u, v\} \in E$ for all $u \neq v \in C$.

Classical MCE seeks cliques that cannot be extended while preserving complete connectivity. In a vector-attributed graph, the attributes add a second, group-level question: how similar are the members represented by a structural clique? We express this semantic condition through two companion quantities.

DEFINITION 3 (PAIRWISE SIMILARITY SUM AND AVERAGE PAIRWISE SIMILARITY). For a clique C of size $k = |C| \geq 2$, the pairwise similarity sum is

$$\text{simSum}(C) = \sum_{\{u,v\} \subseteq C} \text{sim}(u, v),$$

and the average pairwise similarity is

$$\text{avgSim}(C) = \frac{\text{simSum}(C)}{\binom{k}{2}}.$$

DEFINITION 4 (SEMANTIC CLIQUE). Given a threshold $\tau \in [-1, 1]$, a clique C with $|C| \geq 2$ is a τ -semantic clique if $\text{avgSim}(C) \geq \tau$.

For algorithmic purposes, we equivalently track the *semantic surplus*

$$W(C) = \text{simSum}(C) - \tau \binom{k}{2},$$

which is non-negative if and only if $\text{avgSim}(C) \geq \tau$. Working with simSum and W avoids division at every recursive step; avgSim is used only when a human-readable quantity is required.

DEFINITION 5 (MSC). Given $\mathcal{G} = (V, E, \mathbf{X})$ and threshold $\tau \in [-1, 1]$, a τ -semantic clique $C \subseteq V$ is a maximal semantic clique (MSC) if there is no τ -semantic clique $C' \supsetneq C$.

This is inclusion-wise maximality, not merely the absence of a feasible single-vertex extension. The distinction matters because average feasibility is non-monotone. For example, let $\tau = 0.6$ and let four vertices form a clique with $\text{sim}(a, b) = 0.6$, $\text{sim}(a, u) = \text{sim}(b, u) = \text{sim}(a, v) = \text{sim}(b, v) = 0.5$, and $\text{sim}(u, v) = 1$. The feasible clique $\{a, b\}$ has no feasible single-vertex extension, since either three-vertex average is $1.6/3 < 0.6$, but the strict superset $\{a, b, u, v\}$ is feasible with average $3.6/6 = 0.6$. Consequently, algorithms may use the absence of a feasible single extension only to generate candidates; a global containment test is still required for MSC output.

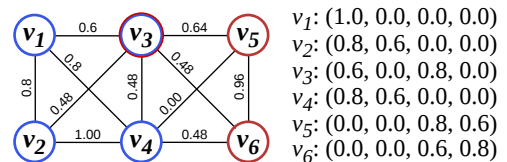


Figure 1: Running example with vector attributes and edge similarities.

EXAMPLE 1. Figure 1 shows a vector-attributed graph with six vertices. Edge labels give the precomputed pairwise similarities. With threshold $\tau = 0.6$, consider the two structural cliques

$$M_1 = \{v_1, v_2, v_3, v_4\}, \quad M_2 = \{v_3, v_4, v_5, v_6\}.$$

Let $C_1 = \{v_1, v_2, v_3, v_4\}$, $C_2 = \{v_3, v_5, v_6\}$, and $C_3 = \{v_3, v_4, v_5, v_6\}$. Their total and average similarities are

$$\begin{aligned} \text{simSum}(C_1) &= 4.16, & \text{avgSim}(C_1) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_2) &= 2.08, & \text{avgSim}(C_2) &\approx 0.693 \geq \tau, \\ \text{simSum}(C_3) &= 2.56, & \text{avgSim}(C_3) &\approx 0.427 < \tau. \end{aligned}$$

Thus C_1 and C_2 are τ -semantic cliques, whereas the structurally valid clique C_3 is not.

Moreover, neither C_1 nor C_2 has a strict superset that is a τ -semantic clique, so both are MSCs.

Problem Statement.

PROBLEM 1 (MSC ENUMERATION). Given a vector-attributed graph $\mathcal{G} = (V, E, X)$ and a threshold $\tau \in [-1, 1]$, enumerate all MSCs of $\mathcal{G}$.

2.2 Challenges

Hardness. MSC enumeration contains classical MCE as a special case. When singleton outputs are excluded and τ does not exceed the minimum similarity on any edge, every structural maximal clique of size at least two satisfies the semantic constraint, so the $\Omega(3^{n/3})$ output lower bound carries over. The associated decision problem is NP-complete.

THEOREM 2.1. Given $\mathcal{G} = (V, E, X)$, threshold $\tau \in [-1, 1]$, and integer k , deciding whether an MSC of size $\geq k$ exists is NP-complete.

PROOF. *NP membership.* The decision instance is positive if and only if it contains a τ -semantic clique C of size at least k . The forward implication is immediate. For the reverse implication, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member, whose size is at least $|C| \geq k$. A certificate therefore need only provide such a feasible clique C : verifying all edges and $\text{avgSim}(C) \geq \tau$ takes $O(|C|^2)$ time. It is unnecessary, and under the non-monotone constraint insufficient, to verify maximality by checking only single-vertex extensions.

NP-hardness. We reduce from the CLIQUE decision problem [20], which asks whether $G_0 = (V, E)$ contains a clique of size $\geq k$. Assign $\mathbf{x}_v = \mathbf{e}_1$ to every vertex, so $\text{sim}(u, v) = 1$ for every edge, and set $\tau = 1$. Under this assignment $\text{avgSim}(C) = 1 \geq \tau$ for every clique C , so every structural maximal clique is an MSC and vice versa. Any clique of size $\geq k$ in G_0 is contained in some structural maximal clique of size $\geq k$, which is also an MSC of size $\geq k$. Conversely, any MSC of size $\geq k$ is a clique of size $\geq k$ in G_0 . Hence a size- k MSC exists if and only if G_0 contains a clique of size $\geq k$, completing the reduction. $\square$

The reduction uses $\tau = 1$ solely to establish a worst-case lower bound for the general MSC decision problem. The quantile protocol in Section 6 studies how non-extremal thresholds shape output size and search behavior in the evaluated graphs.

The output sets of MSC enumeration and MCE are incomparable: a structural maximal clique with $\text{avgSim}(M) < \tau$ may contain

semantically maximal subsets that MCE never generates, while valid MSCs need not be structurally maximal. Therefore, enumerating structural maximal cliques and retaining only those that pass the average constraint is incomplete, rather than merely inefficient. The output size, and hence the computational cost, is τ -dependent in a non-trivial way that we characterize below.

Output Distribution and Practical Scope. The number of MSCs and their sizes vary substantially with τ . At sufficiently low τ , almost every clique satisfies the average constraint and the output converges toward that of classical MCE. As τ increases, large structural cliques that previously satisfied the constraint cease to do so and are replaced by many smaller semantically feasible sub-cliques, none of which subsumes another. This *fragmentation effect* can inflate the output size by orders of magnitude relative to MCE. At sufficiently high τ , the constraint eliminates most cliques outright and the output shrinks considerably.

The intermediate fragmentation band carries the largest output volume and is therefore inherently time-consuming—any correct algorithm must enumerate all output cliques, so the output size itself imposes a lower bound on runtime. Low τ is appropriate when the goal is to filter out structurally maximal cliques that are semantically too loose; high τ is appropriate when the goal is to surface only the most semantically coherent groups. The exact τ values at which these transitions occur depend on the similarity distribution of the underlying dataset and cannot be fixed a priori. Section 6 therefore evaluates a fixed grid of per-dataset edge-similarity quantiles rather than choosing absolute thresholds after observing runtimes.

Non-monotonicity and Framework Incompatibility. The core difficulty is that avgSim is non-monotone over the subset lattice. In classical MCE, structural feasibility is monotone and maximality is enforced implicitly by the $P = \emptyset$ invariant at every leaf, so the set of feasible extensions can be characterized solely by the current state of C . Under the average constraint, no such invariant exists: a partial clique with $\text{avgSim}(C) < \tau$ may still grow into a valid MSC, while one with $\text{avgSim}(C) \geq \tau$ may admit no feasible extension. Furthermore, whether a candidate vertex can participate in a valid extension depends on its accumulated similarity to the current partial clique, which evolves throughout the recursion and cannot be captured by structural conditions alone. As a result, the pruning mechanisms based on structural monotonicity lose their foundation, and the pivot strategy no longer applies. A direct BK adaptation must therefore retain structurally valid intermediate states even when they are currently infeasible. In the baseline developed next, τ validates terminal semantic candidates but does not reduce the structural recursion itself.

3 FRAMEWORK

Framework Overview. We derive SEMBK, a direct baseline that adapts the Bron–Kerbosch (BK) recursion to the semantic-clique setting under the average constraint. The standard partition (P, X) remains useful for structural enumeration, but the semantic constraint is defined at the clique level: whether a partial clique can eventually satisfy the threshold depends on avgSim over all vertex pairs within the clique. As a consequence, the pivoting strategy, which relies on local structural properties, is no longer directly

Algorithm 1: Semantic BK with Check

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  and degeneracy ordering  $\sigma$ ;
4 Each worker initializes a private result set  $\mathcal{R}_t \leftarrow \emptyset$ ;
5 for each  $v \in \sigma$  in parallel do
6    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
7    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
8   Initialize  $\text{acc}[u] \leftarrow \text{sim}(u, v)$  for all  $u \in N(v)$ ;
9   Search( $C = \{v\}, s_C = 0, P, X, \text{acc}, \text{mode} = \text{ENUM}, 0$ )
10 return  $\bigcup_t \mathcal{R}_t$ ;

11 Function Search( $C, s_C, P, X, \text{acc}, \text{mode}, k_{\min}$ ):
12    $k \leftarrow |C|$ ,  $\text{hasFeasibleDesc} \leftarrow \text{false}$ ;
13   for each  $v \in P$  do
14      $s_D \leftarrow s_C + \text{acc}[v]$ ;
15      $P \leftarrow P \setminus \{v\}, P' \leftarrow P \cap N(v), X' \leftarrow X \cap N(v)$ ;
16      $\text{acc}[u] \leftarrow \text{acc}[u] + \text{sim}(u, v)$  for  $u \in P' \cup X'$ ;
17      $\text{res} \leftarrow \text{Search}(C \cup \{v\}, s_D, P', X', \text{acc}, \text{mode}, k_{\min})$ ;
18      $\text{acc}[u] \leftarrow \text{acc}[u] - \text{sim}(u, v)$  for  $u \in P' \cup X'$ ;
19     if  $\text{res}$  then
20        $\text{hasFeasibleDesc} \leftarrow \text{true}$ ;
21       if  $\text{mode} = \text{CHECK}$  then return true;
22    $X \leftarrow X \cup \{v\}$ ;
23   if  $\neg \text{hasFeasibleDesc}$  then
24     if  $k \geq 2 \wedge s_C < \tau \binom{k}{2}$  then return false;
25     if  $\text{mode} = \text{CHECK}$  then return  $k > k_{\min}$ ;
26     if  $X \neq \emptyset$  then
27       if Search( $C, s_C, X, \emptyset, \text{acc}, \text{CHECK}, k$ ) then
28         return true;
29     if  $k \geq 2$  then  $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup \{C\}$ ;
30     return true;
31   return  $\text{hasFeasibleDesc}$ ;

```

applicable. The baseline therefore retains the BK recursion and degeneracy ordering while excluding pivoting, and enforces semantic maximality via a secondary CHECK pass on X at each leaf.

Figure 2 shows both terminal decisions. After emitting C_1 , the same root later reaches the feasible set $D = \{v_1, v_2, v_4\}$; its exclusion set contains v_3 , so CHECK reconstructs the feasible strict superset C_1 and suppresses D . In the second root, the infeasible intermediate $\{v_3, v_4, v_5\}$ is not pruned while v_6 remains in P . Its terminal descendant C_3 also fails τ , whereas the sibling C_2 is feasible and survives CHECK because the only tested strict superset is C_3 .

Degeneracy-Root Ownership. The outer loop iterates over vertices in degeneracy order σ . For a seed vertex v , its later neighbors under σ form the candidate set P , and its earlier neighbors are placed in X . This partition guarantees that every structural maximal clique is generated by exactly one seed, because the unique vertex of minimum degeneracy index in any clique serves as its seed and sees all co-members in its P . Since every MSC is contained in at least one structural maximal clique (as established later in Lemma 4.1), the same partition ensures completeness for the semantic problem without modification.

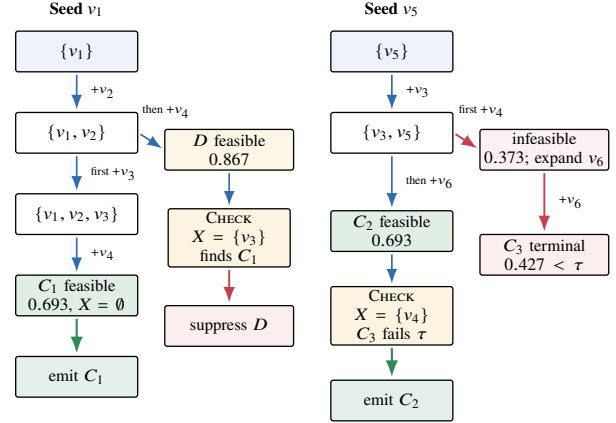


Figure 2: Selected SEMBK branches on the running example.

Terminal Feasibility and Maximality Check. Within each recursive call, the algorithm explores all vertices in P as expansion roots. For each candidate v , it updates $s_C = \text{simSum}(C)$ and the auxiliary accumulation array acc , then recurses on the reduced candidate and exclusion sets (P', X') . The Boolean return value records whether the subtree contains a feasible terminal. When no recursive child reports one ($\text{hasFeasibleDesc} = \text{false}$), the algorithm first tests the average constraint: if $\text{avgSim}(C) < \tau$, the branch is pruned immediately. Otherwise, semantic maximality must be verified: a clique C with $\text{avgSim}(C) \geq \tau$ and $P = \emptyset$ could still be non-maximal if some vertex in X can extend it to a larger valid clique. The algorithm therefore invokes SEARCH on X in CHECK mode with $k_{\min} = |C|$. In CHECK mode, the recursion explores extensions through X and returns *true* as soon as it finds any valid clique strictly larger than C ; if no such extension exists, C is emitted. The strict size condition $k > k_{\min}$ is necessary because the CHECK recursion may reach sub-cliques of C itself; without recording $k_{\min}$, these would be misidentified as valid extensions.

Why pivoting is not used. In standard BK, a pivot $p \in P \cup X$ guarantees that each structural maximal clique in the current branch contains either p or a vertex in $P \setminus N(p)$. Hence vertices in $P \cap N(p)$ need not become expansion roots. This argument depends on structural feasibility being preserved when an adjacent pivot is added.

The average constraint invalidates that implication. Consider a complete graph on $\{s, p, u, w\}$ at the BK state $C = \{s\}, P = \{p, u, w\}$, and $X = \emptyset$. Let the three similarities within $\{s, u, w\}$ equal 0.9, the three similarities incident to p equal 0, and $\tau = 0.6$. Then $\{s, u, w\}$ is an MSC with average 0.9, whereas its only strict clique superset has average 0.45. Choosing p as the pivot leaves only p in $P \setminus N(p)$, so every explored child contains p and the valid MSC $\{s, u, w\}$ is lost. Pivot-free branching is therefore required for the direct semantic BK baseline. **Complexity.**

THEOREM 3.1 (TIME AND SPACE COMPLEXITY). Let d_G be the degeneracy of G , let $\mathcal{L}$ be the feasible terminal states reached by the outer search, and let X_C be the candidate set examined by CHECK for $C \in \mathcal{L}$. The running time is $O^*(n2^{d_G} + \sum_{C \in \mathcal{L}} 2^{|X_C|})$, where O^*

suppresses polynomial set-intersection costs. A graph-independent upper bound is $O^*(3^n)$. With T workers, the implementation uses $O(m + nD + Tn^2)$ worst-case space.

PROOF. A degeneracy seed has at most d_G later neighbors. Without pivoting, its outer recursion may visit every subset of those neighbors, giving $O^*(n2^{d_G})$ states. For a feasible terminal C , CHECK may independently visit every subset of X_C . The same earlier extension can be reconsidered for several terminals; charging a state to a pair of nested vertex sets gives the safe $O^*(3^n)$ bound. Similarities and sparse adjacency require $O(m + nD)$ storage. Each worker owns an $O(n)$ accumulator, while decreasing candidate arrays retained by a worst-case CHECK stack total $O(n^2)$. $\square$

For comparison, degeneracy-oriented pivoted MCE has the time bound [11, 12]

$$O(d_G n 3^{d_G/3})$$

The average constraint introduces more than an $O(|C|^2)$ validation cost: valid MSCs may occur at nonmaximal structural states, so the direct search may inspect all 2^{d_G} subsets below a seed and revisit earlier-neighbor subsets in CHECK. This gap formalizes the additional search complexity relative to structural MCE.

Design Implication. SEMBK follows the problem definition directly and requires only local search state: degeneracy roots assign structural subspaces, incremental accumulators maintain similarity sums, and CHECK avoids a global visited table. It is therefore a transparent same-definition baseline. Its limitations are also fundamental to this organization. Because infeasible intermediate cliques may have feasible supersets, τ does not prune the main recursion; because pivoting is unsafe, the search loses BK’s principal structural reduction; and the CHECK pass may revisit earlier-neighbor subspaces for multiple terminals. Section 4 addresses these costs with two search organizations tailored to different threshold regimes.

4 PROPOSED ALGORITHMS

This section presents two exact algorithms that exploit the global structure of the average constraint from different directions.

STRSUB (Section 4.1) resolves the pivot incompatibility by decoupling structural and semantic search. The pivot is applied without modification to a τ -agnostic structural phase, restoring the $O(3^{n/3})$ BK guarantee; τ is then enforced in a subsequent intra-clique enumeration phase. Complementary sorted-edge and vertex-contribution bounds skip subset-size layers that cannot satisfy τ . Its cost is governed by structural MCE and by the sizes of the resulting host cliques.

MONOSEM MCE (Section 4.2) departs from the BK framework entirely. Motivated by the observation that STRSUB, despite its pruning improvements, still processes all structural maximal cliques irrespective of their semantic quality, MONOSEM MCE introduces a canonical reverse-search strategy in which τ drives enumeration from the first expansion step. A deterministic feasible parent gives each feasible clique one generation path, while depth-first traversal avoids storing a complete size layer. The method is particularly effective at high τ , where few candidate extensions pass the surplus test.

Graph size and structural clique growth determine the crossover between the two methods. We therefore evaluate them on the same

quantile grid: STRSUB provides broad threshold coverage, while MONOSEM MCE specializes the search for selective high- τ workloads.

4.1 STRSUB: Structural Decomposition and Subset Enumeration

Structural–Semantic Decoupling and Containment. The validity of the Bron–Kerbosch pivot rests on a structural invariant: every structural maximal clique must contain at least one vertex from $P \setminus N(u)$ for any pivot $u \in P \cup X$. This invariant is violated when the average-similarity constraint is incorporated directly into the branching procedure, because avgSim is a *global* property of a candidate set and does not decompose along the local neighborhood structure used to justify the pivot. Consequently, the semantic variant SEMBK must forgo pivoting entirely, which undermines the $O(3^{n/3})$ worst-case guarantee of the standard algorithm [32].

For a host clique M and target size k , let $\bar{e}_k$ be the mean of the $\binom{k}{2}$ largest pairwise similarities in M . For each $v \in M$, let $q_v(k)$ be the sum of its $k - 1$ largest incident similarities within M , and let $\bar{q}_k$ be the sum of the k largest $q_v(k)$ values divided by $k(k - 1)$. Both quantities upper-bound the average similarity of any size- k subset; Properties 4.1 and 4.2 prove these bounds below.

Figure 3 separates the two host-clique outcomes. Gray marks structural or global processing, green marks accepted feasible states, and amber marks the bounded search triggered by an infeasible host. Thus M_1 is accepted as C_1 , whereas subset enumeration within M_2 recovers C_2 .

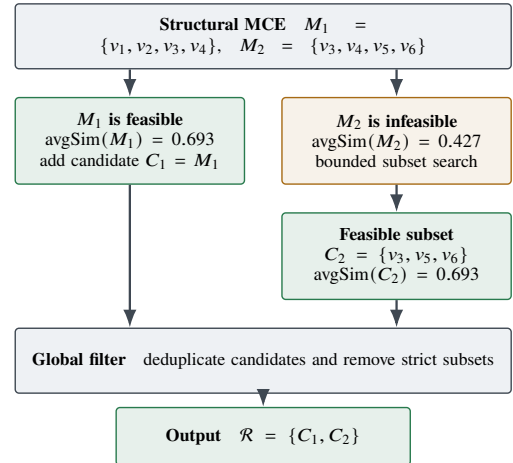


Figure 3: STRSUB execution on the running example.

The key insight behind STRSUB is that the structural and semantic layers of the problem can be *decoupled*. Specifically, the Tomita pivot can be applied without modification to the purely structural problem of enumerating maximal cliques in G , because that problem carries no average-similarity constraint. The threshold τ is then enforced in a second, independent pass confined to each structural maximal clique. The theoretical basis for this decoupling is the following containment lemma.

Algorithm 2: STRSUB: Structural Decomposition and Subset Enumeration

Input: $G = (V, E)$, similarities $\{\text{sim}(u, v)\}_{(u,v) \in E}$, threshold τ
Output: All MSCs $\mathcal{R}$

```

1  $\mathcal{F} \leftarrow \emptyset$ ;
  /* Pipelined structural MCE and semantic processing */
2 Build degeneracy ordering  $\sigma$ ;
3 for each  $v \in \sigma$  in parallel do
4    $P \leftarrow \{u \in N(v) \mid \sigma(u) > \sigma(v)\}$ ;
5    $X \leftarrow \{u \in N(v) \mid \sigma(u) < \sigma(v)\}$ ;
6   Run BK( $\{v\}, P, X$ ) with the following callback at each leaf  $M$ ;
  /* Callback: intra-clique semantic enumeration */
7 for each structural leaf  $M$  produced by a worker do
8   if  $\text{avgSim}(M) \geq \tau$  then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{M\}$ ; continue;
  /* Admissible layer bounds */
9   Sort edges of  $M$  by similarity descending;
10   $k^* \leftarrow \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\}$ ;
11  if  $k^*$  does not exist then continue;
12   $lmax \leftarrow \emptyset$ ;  $umask \leftarrow 0$ ;
13  for  $k \leftarrow k^*$  to 2 do
14    Compute the vertex-contribution bound  $\bar{q}_k$  for size  $k$ ;
15    if  $\bar{q}_k < \tau$  then continue;
16    for each size- $k$  subset  $S \subseteq M$  do
17      if  $(S \& umask) = S$  and  $\exists T \in lmax : S \subseteq T$  then
18        continue;
19      if  $\text{avgSim}(S) \geq \tau$  then
20         $lmax \leftarrow lmax \cup \{S\}$ ;
21         $umask \leftarrow umask \mid S$ ;
22         $\mathcal{F} \leftarrow \mathcal{F} \cup \{S\}$ ;
  /* Phase 3: Exact global filter */
22 Sort  $\mathcal{F}$  by  $|\cdot|$  descending; remove exact duplicates;
23 Initialize an empty inverted index  $inv : v \mapsto \emptyset$ ;
24  $\mathcal{R} \leftarrow \emptyset$ ;
25 for each equal-size group  $\mathcal{G} \subseteq \mathcal{F}$  in descending size do
26   for each  $S \in \mathcal{G}$  in parallel do
27      $r \leftarrow \arg \min_{v \in S} |inv[v]|$ ;
28     Keep  $S$  iff no  $T$  referenced by  $inv[r]$  satisfies  $S \subsetneq T$ ;
29   Insert every kept  $S \in \mathcal{G}$  into  $\mathcal{R}$  and  $inv$ ;
30 return  $\mathcal{R}$ ;
```

LEMMA 4.1 (STRUCTURAL SUPERSET OF MSCs). *Every MSC is contained in at least one structural maximal clique of G . That is, for every MSC C , there exists a structural maximal clique M such that $C \subseteq M$.*

PROOF. By definition, every MSC is a clique of G . Let C be an arbitrary MSC, and consider the family

$$\mathcal{F} = \{D \subseteq V \mid C \subseteq D \text{ and } D \text{ is a clique in } G\}.$$

Since $C \in \mathcal{F}$, the family is nonempty. Because V is finite, $\mathcal{F}$ is finite, so we may choose a set $M \in \mathcal{F}$ that is maximal under inclusion. By construction, M is a clique containing C , and no vertex $w \in V \setminus M$ extends M to a larger clique; hence M is a structural maximal clique of G , and $C \subseteq M$. $\square$

Lemma 4.1 establishes that the complete collection of MSCs can be recovered by (i) enumerating all structural maximal cliques, and (ii) searching within each structural maximal clique for semantically feasible maximal subsets. Step (i) is a purely structural problem to which the full Tomita pivot applies, restoring the $O(3^{n/3})$ worst-case bound of standard BK. The threshold τ enters only in step (ii), where the search space is bounded by the size of a single structural clique. Algorithm 2 implements this strategy in three phases.

Structural Host Enumeration. Phase 1 enumerates all structural maximal cliques of G using the standard BK algorithm equipped with the Tomita pivot [32] and degeneracy ordering [12]. Vertices are processed in degeneracy order σ ; for each seed vertex v , the candidate set P and exclusion set X are initialized to the forward and backward neighborhoods of v in σ , respectively, and recursive BK is invoked on the triple $(\{v\}, P, X)$. Seeds are dispatched in parallel via OpenMP dynamic scheduling. Crucially, this phase is entirely τ -agnostic: the threshold plays no role in branching decisions, so the full pivot strategy and its associated complexity bound are preserved. The implementation does not materialize the set of all structural maximal cliques: when BK reaches a leaf M , the same worker immediately invokes the Phase 2 callback and retains only its thread-local semantic candidates.

Bounded Semantic Subset Search. For each structural maximal clique M of size m produced by Phase 1, Phase 2 enumerates all semantically maximal subsets of M with respect to τ . Subsets of M are generated in descending order of cardinality using native-word bitmasks when possible and dynamic bitsets for larger host cliques; no mask-width limit is imposed. Pairwise similarities are precomputed into an $m \times m$ local matrix to amortize repeated avgSim evaluations.

A *local maximality filter* prunes redundant candidates during enumeration. The filter maintains a set $lmax$ of valid subsets discovered so far within M , together with a union mask $umask = \bigcup_{T \in lmax} T$ that provides a constant-time necessary condition for subsumption. A candidate subset S is skipped if $S \subseteq T$ for some $T \in lmax$, verified first via the bitmask test $(S \& umask) = S$ and then by a linear scan over $lmax$. This filter is correct because, if $S \subsetneq T$ and $\text{avgSim}(T) \geq \tau$, then S cannot be semantically maximal in G regardless of its own average similarity. Structural leaves are processed by their owning workers. The candidate size layers are further reduced by the admissible bounds described below.

Global Maximality Reconciliation. An MSC may occur in multiple structural maximal cliques, so semantic subset search may emit duplicate or subsumed candidates. Global reconciliation consolidates these candidates into the final result set $\mathcal{R}$.

The phase proceeds in two steps. First, candidates are sorted by decreasing size with a lexicographic tie break, and exact duplicates are removed in $O(N \log N)$ time, where $N = |\mathcal{F}|$. Second, equal-size groups are processed in that order. The inverted index $inv : v \mapsto \{T \in \mathcal{R} \mid v \in T\}$ contains only retained strict supersets of the current group. For each candidate S , the algorithm identifies its *rarest vertex* $r = \arg \min_{v \in S} |inv[v]|$ and scans only $inv[r]$; a sorted-merge traversal determines in $O(|S| + |T|)$ time whether $S \subsetneq T$. Queries within one group read the same frozen index and run in parallel, after which the retained candidates are inserted. Equal-size sets cannot strictly contain one another. Moreover, if a

discarded larger candidate contains S , transitivity guarantees that a retained candidate containing both will be indexed. The grouped filter is therefore equivalent to testing all larger candidates while avoiding their indexing and synchronization costs.

Admissible Size-Layer Bounds. Phase 2 as described iterates over all subset sizes from $|M| - 1$ down to 2. A key observation allows the starting size to be reduced: the best achievable avgSim for any size- k subset of M is bounded above by the mean of the $\binom{k}{2}$ largest pairwise similarities in M .

PROPERTY 4.1 (SORTED-EDGE UPPER BOUND). *Let $e_1 \geq e_2 \geq \dots \geq e_{\binom{m}{2}}$ be the pairwise similarities of M sorted in descending order. For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{e}_k := \frac{1}{\binom{k}{2}} \sum_{i=1}^{\binom{k}{2}} e_i.$$

PROOF. Any size- k subset $S \subseteq M$ uses exactly $\binom{k}{2}$ pairs, each contributing at most the corresponding value in the sorted sequence $e_1 \geq e_2 \geq \dots$; hence $\text{simSum}(S) \leq \sum_{i=1}^{\binom{k}{2}} e_i$, and dividing both sides by $\binom{k}{2}$ gives the result. $\square$

The sorted-edge bound ignores that its largest edges may be concentrated on too few vertices to coexist in one size- k subset. A complementary bound tracks which vertices support those similarities. For each $v \in M$, let $q_v(k)$ be the sum of the $k - 1$ largest similarities between v and vertices in $M \setminus \{v\}$. Let $q_{(1)}(k) \geq \dots \geq q_{(m)}(k)$ denote these values in descending order.

PROPERTY 4.2 (VERTEX-CONTRIBUTION UPPER BOUND). *For any size- k subset $S \subseteq M$,*

$$\text{avgSim}(S) \leq \bar{q}_k := \frac{1}{k(k-1)} \sum_{i=1}^k q_{(i)}(k).$$

PROOF. For every $v \in S$, the sum of its similarities to the other vertices of S uses $k - 1$ vertices from $M \setminus \{v\}$ and is therefore at most $q_v(k)$. Summing over S counts each internal pair twice, so

$$2\text{simSum}(S) \leq \sum_{v \in S} q_v(k) \leq \sum_{i=1}^k q_{(i)}(k).$$

Division by $k(k-1)$ gives the claimed average-similarity bound. $\square$

Note that $\bar{e}_k$ is non-increasing in k : a larger subset requires more edges, drawing further into the sorted list and lowering the mean. Before enumeration, all pairwise similarities of M are sorted once in $O(m^2 \log m)$ and prefix means $\bar{e}_k$ are computed for all k . The algorithm identifies

$$k^* = \max\{k \in [2, |M| - 1] \mid \bar{e}_k \geq \tau\},$$

and begins enumeration at $k = k^*$ rather than $|M| - 1$. Because $\bar{e}_k$ is non-increasing, all sizes $k > k^*$ satisfy $\bar{e}_k < \tau$, certifying that no subset of those sizes can meet the constraint; they are skipped entirely. If no such k^* exists, the whole clique M is skipped. For every remaining size k , the algorithm also skips the layer when $\bar{q}_k < \tau$. The two tests are independently admissible, so using their minimum cannot remove a feasible subset. Computing all per-vertex

similarity lists and prefix sums costs $O(m^2 \log m)$, matching the sorting order of the sorted-edge preprocessing.

Applicable Regime. STRSUB applies Tomita pivoting during structural MCE, whose cost is $O(d_G n 3^{d_G/3})$ and independent of τ . At low thresholds, large feasible subsets are identified early within each structural host. At higher thresholds, the sorted-edge and vertex-contribution bounds skip infeasible size layers. Structural MCE remains a fixed front-end cost, which motivates the canonical search developed next for selective thresholds.

Complexity.

THEOREM 4.1 (TIME COMPLEXITY OF STRSUB). *Let $\text{mathcal{M}}$ denote the family of all structural maximal cliques of G . For each M*

in

mathcal{M}, let Q_M denote the number of exact local containment probes and let $b_M =$

lceil |M|/w

be its bitmap length in machine words. STRSUB runs in

$$O\left(d_G n 3^{d_G/3} + \sum_{M \in \mathcal{M}} \left(2^{|M|} |M|^2 + Q_M b_M\right) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right)$$

time, where d_G is the degeneracy of G , $N = |\mathcal{F}|$ is the number of candidates before global deduplication, $K = \max_{S \in \mathcal{F}} |S|$, and $L_{\min}(S) = \min_{v \in S} |\text{inv}[v]|$.

PROOF. The structural phase uses degeneracy-oriented Tomita BK [12], with cost $O(d_G n 3^{d_G/3})$. For a host M of size h , subset enumeration visits at most 2^h subsets and evaluates each from the cached similarity matrix in $O(h^2)$ time. Exact local maximality performs Q_M bitmap containment probes of b_M words each; keeping this term explicit also covers instances with many feasible local subsets. The final filter sorts N candidates in $O(KN \log N)$ time. It then scans the shortest inverted list for each S , with at most $L_{\min}(S)$ sorted-set containment checks of $O(K)$ time. $\square$

The first term matches sparse structural MCE. The additional $2^{|M|} |M|^2$ term is specific to MSC enumeration: because feasibility is non-monotone, a structurally maximal host M may contain exponentially many semantic candidates at different sizes. The two admissible bounds remove certifiably infeasible size layers in practice but do not change this worst case.

Remark. Without pruning, the enumeration term is exponential in each structural host, and Q_M is output sensitive. The admissible bounds replace $2^{|M|}$ by $\sum_{k=2}^{k^*} \binom{|M|}{k}$ when all larger layers are certified infeasible; local containment reduces the candidates reaching the global filter without changing the output set.

4.2 MONOSEMMCE: High- τ Canonical Reverse Search

Motivation and Core Insight. Even with the size-layer bounds, STRSUB is bound to Phase 1: it enumerates all structural maximal cliques before any semantic evaluation takes place. In the high- τ regime, the vast majority of structural cliques may be semantically barren, yet each still contributes Phase 1 cost. More fundamentally, the BK framework offers no mechanism to incorporate τ into branching decisions—the threshold appears only in validation,

never in the search logic itself. This structural overhead motivates abandoning the BK skeleton entirely.

The obstacle to integrating τ directly into search is the non-monotonicity of avgSim : an infeasible clique can have a feasible strict superset. A search that retains only feasible states is therefore complete only if every feasible target has a feasible construction path. Theorem 4.2 supplies such a path through the deletion rule defined next. The resulting proof has two enumeration consequences: feasible parenthood gives a path from every target to an edge root, establishing completeness, while deterministic tie-breaking gives each non-root state one parent, establishing unique generation.

For a vertex v in a clique C , define its *similarity contribution* as $a_C(v) = \sum_{u \in C \setminus \{v\}} \text{sim}(u, v)$: the sum of its similarities to the other vertices in C . We delete a vertex with minimum similarity contribution, breaking ties by vertex identifier. The theorem below proves that the remaining clique is feasible. This deterministic rule gives every feasible clique a unique parent, called its *canonical parent*. Reverse search accepts an extension only when its parent under this rule is the current clique. Thus every feasible clique is generated once without a visited table or per-layer deduplication.

The implementation traverses each canonical tree by depth-first search (DFS) and parallelizes its feasible edge roots. It rejects an extension whenever its updated surplus is negative. Unlike the previous level-synchronous algorithm, it stores no complete size layer; working memory consists of the graph representation, thread-local accumulators, DFS scratch space, and output candidates. A final containment filter is still necessary, because absence of a feasible single extension is weaker than inclusion-wise maximality.

Feasible Parents and the Canonical Forest.

THEOREM 4.2 (FEASIBLE MINIMUM-CONTRIBUTION DELETION). *Let C be a τ -semantic clique of size $k \geq 3$, with the similarity contribution $a_C(v)$ defined above. If $d(C)$ minimizes the pair $(a_C(v), v)$ lexicographically, then $C \setminus \{d(C)\}$ is a τ -semantic clique. Repeated application of this rule therefore reaches a feasible edge, and reversing these deletions gives a non-increasing-average insertion path.*

PROOF. Each pair is counted once at each endpoint. The minimum contribution therefore satisfies $a_C(d(C)) \leq 2\text{simSum}(C)/k$. Deleting $d(C)$ leaves similarity sum at least $\text{simSum}(C)(k-2)/k$, and hence

$$\text{avgSim}(C \setminus \{d(C)\}) \geq \frac{\text{simSum}(C)(k-2)/k}{\binom{k-1}{2}} = \text{avgSim}(C) \geq \tau.$$

Deterministic tie-breaking makes $d(C)$ unique. Applying the same argument until two vertices remain preserves feasibility at every deletion; reversing the sequence proves the insertion-path claim. $\square$

Define $p(C) = C \setminus \{d(C)\}$ for every feasible clique C of size at least three. The theorem makes $p(C)$ a feasible and unique parent, so all feasible cliques form a forest rooted at feasible edges. This property does not make arbitrary insertions or deletions monotone; it identifies one complete feasible path for every target. MONO-SEM-MCE traverses precisely these owned parent-child edges and rejects alternative constructions before recursion.

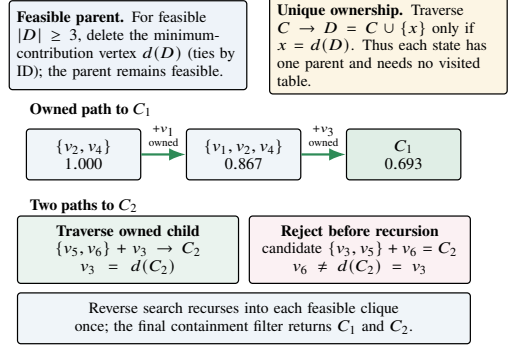


Figure 4: Canonical ownership on the running example.

Algorithm 3: MONOSEM-MCE: Canonical Reverse Search

```

1 Input:  $G = (V, E)$ , semantic vectors  $\{x_v\}$ , threshold  $\tau$ 
2 Output: All MSCs  $\mathcal{R}$ 
3 Precompute  $\text{sim}(u, v)$  for all  $(u, v) \in E$ ;
4  $\mathcal{F} \leftarrow \emptyset$ ;
5 for each  $(u, v) \in E$  with  $u < v$  and  $\text{sim}(u, v) \geq \tau$  in parallel do
6    $C \leftarrow \{u, v\}$ ;  $s_C \leftarrow \text{sim}(u, v)$ ;
7    $P \leftarrow N(u) \cap N(v)$ ;
8    $\text{acc}[w] \leftarrow \text{sim}(w, u) + \text{sim}(w, v)$  for  $w \in P$ ;
9    $a_C(u) = a_C(v) \leftarrow \text{sim}(u, v)$ ;
10  Search( $C, s_C, P, \text{acc}, a_C$ );
11 Apply Algorithm 2's grouped strict-containment filter to  $\mathcal{F}$ ,
    obtaining  $\mathcal{R}$ ;
12 return  $\mathcal{R}$ ;

13 Function Search( $C, s_C, P, \text{acc}, a_C$ ):
14    $k \leftarrow |C|$ ;  $w_C \leftarrow s_C - \tau \binom{k}{2}$ ;
15   hasValid  $\leftarrow$  false;
16   for each  $v \in P$  do
17      $\delta \leftarrow \text{acc}[v]$ ;
18     if  $w_C + \delta - \tau k < 0$  then continue;
19     hasValid  $\leftarrow$  true;
20      $D \leftarrow C \cup \{v\}$ ;
21     Update  $a_D$  from  $a_C$  and  $\text{sim}(v, u)$  for  $u \in C$ ;
22     if  $v \neq \arg \min_{x \in D} (a_D(x), x)$  then continue;
23      $P' \leftarrow P \cap N(v)$ ;
24      $\text{acc}'[w] \leftarrow \text{acc}[w] + \text{sim}(w, v)$  for  $w \in P'$ ;
25     Search( $D, s_C + \delta, P', \text{acc}', a_D$ );
26   if not hasValid then  $\mathcal{F} \leftarrow \mathcal{F} \cup \{C\}$ ;
  
```

Figure 4 applies this rule to the running example. The green paths insert the canonical deletion of each child, while the red path forms C_2 as a non-owned candidate and rejects it before recursion.

Feasible Edge Roots. Every feasible clique of size at least two reaches a feasible edge under the deterministic deletion rule, so Algorithm 3 starts one tree for each edge with similarity at least τ . A state stores the clique C , its pairwise sum $s_C = \text{simSum}(C)$, the common-neighbor set P , candidate accumulators acc , and the

member contributions a_C . No BK exclusion set is required: uniqueness follows from the canonical-parent test rather than from an insertion order.

Surplus and Canonical-Child Tests. For $k = |C|$ and $\delta = \text{acc}[v]$, the extension satisfies

$$W(C \cup \{v\}) = W(C) + \delta - \tau k.$$

If this value is negative, the child itself is infeasible and cannot lie on the canonical path of any feasible answer, so it is safely skipped. For a feasible child $D = C \cup \{v\}$, the algorithm updates every member’s similarity contribution and accepts D only if $v = d(D)$. Equivalently, the canonical parent of D must be exactly C . This test removes alternative insertion paths before recursion rather than after state materialization. The flag *hasValid* is set before the ownership test because it records the existence of any feasible one-vertex extension, not the existence of an owned child. A state with such an extension cannot itself be inclusion-wise maximal; the extension is generated from its unique canonical parent.

The path property is used to assign unique parent ownership rather than to order candidates. The algorithm tests child feasibility directly and then applies the minimum-contribution deletion rule. This removes duplicate paths before recursion without a candidate-order cutoff. Accumulator values are restored from saved values when backtracking, so the result does not depend on reversing a sequence of floating-point additions.

Inclusion-Wise Maximality. A state with no feasible single-vertex extension is added to $\mathcal{F}$, but is not yet declared an MSC. If such a candidate C is not inclusion-wise maximal, the finite family of feasible clique supersets of C contains an inclusion-wise maximal member M . The canonical search enumerates M exactly once, and M has no feasible single extension, so it also belongs to $\mathcal{F}$. The final grouped inverted-index containment filter consequently removes C . Conversely, an MSC is not contained in any other feasible candidate and is retained.

THEOREM 4.3 (ENUMERATION CORRECTNESS). *Algorithm 3 generates every τ -semantic clique of size at least two exactly once and returns exactly all MSCs.*

PROOF. Existence follows by reversing the repeated deterministic deletions in Theorem 4.2; every prefix is feasible and therefore passes the surplus test. Uniqueness follows because every non-root feasible clique has the single parent $C \setminus \{d(C)\}$, and only that parent accepts it as a child. The preceding containment argument proves that the final filter retains precisely the inclusion-wise maximal feasible cliques. $\square$

Complexity.

THEOREM 4.4 (TIME AND SPACE COMPLEXITY OF MONOSEMMCE). *Let F_τ be the number of feasible cliques of size at least two, K the maximum clique size, Δ the maximum degree, T the number of workers, and $N = |\mathcal{F}|$ the number of terminal candidates. A conservative bound for canonical search is*

$$O\left(F_\tau \Delta (\Delta + K \log \Delta) + KN \log N + K \sum_{S \in \mathcal{F}} L_{\min}(S)\right),$$

with $O(m + T(n + K\Delta) + NK)$ words of space in addition to vector storage.

PROOF. Canonical ownership bounds accepted states by F_τ . A state tests at most Δ candidates. For each candidate, common-neighbor intersection and accumulator maintenance cost $O(\Delta)$ conservatively, while updating and comparing at most K member contributions uses cached similarities with $O(\log \Delta)$ lookup. This is bounded by $O(\Delta(\Delta + K \log \Delta))$ per accepted state. The final filter has the same $O(KN \log N + K \sum_S L_{\min}(S))$ cost as in STRSUB.

Sparse similarity adjacency and neighbor bitmaps use $O(m)$ words. Each worker owns an $O(n)$ generation-tagged accumulator and $O(K\Delta)$ DFS scratch space. Terminal candidates occupy $O(NK)$ words; no feasible-size layer or visited-state table is materialized. $\square$

Unlike structural MCE, whose search is organized around maximal structural cliques, canonical search visits the F_τ feasible cliques induced by the semantic threshold. This family can substantially exceed the structural output at intermediate thresholds, but contracts sharply when τ is selective.

Remark. Increasing τ contracts the feasible-state family F_τ , which gives MONOSEMMCE its selective-threshold operating region. STRSUB instead pays the structural MCE cost and is less sensitive to F_τ .

5 IMPLEMENTATION

All methods use a common C++17 graph and similarity engine so that the measured differences reflect their search organizations. The engine first applies a degree-descending breadth-first reorder to improve locality, while maintaining the original-to-internal identifier map used by every enumerator. Adjacency rows are represented by *sparse blocked bitmaps*: only nonempty 64-bit blocks are stored, and intersections merge block identifiers before applying word-level AND operations. This layout retains bitmap operations in dense local regions without allocating an n -bit row per vertex. Short exclusion lists in SEMBK are scanned directly; longer lists are converted to the same blocked form.

Similarity Representation. Input vectors are normalized once and stored in 64-byte-aligned float32 arrays. Advanced Vector Extensions 2 (AVX2) kernels compute dot products, after which each edge similarity is cached with the sparse adjacency. Similarity sums, candidate accumulators, semantic surplus, and threshold products use double precision. All algorithms apply the same inclusive comparison $\text{simSum}(C) \geq \tau \binom{|C|}{2}$ and no method-specific tolerance. Within a structural host M , STRSUB materializes an $|M| \times |M|$ float32 matrix so that repeated subset tests avoid graph-level similarity lookups. The end-to-end timer starts before the initialization and preprocessing steps described above, so reported runtimes include edge-similarity computation, adjacency and cache construction, search, and result consolidation; phase counters isolate search work when we analyze individual mechanisms.

Incremental Search State. For each candidate w , the algorithms maintain $\text{acc}[w] = \sum_{u \in C} \text{sim}(u, w)$. Extending C by v updates only the surviving candidates as $\text{acc}'[w] = \text{acc}[w] + \text{sim}(w, v)$, reducing an $O(|P'| |C|)$ recomputation to $O(|P'|)$. Each worker owns a generation-tagged array, so starting a new root performs a logical

reset rather than clearing $O(n)$ entries. SEMBK and STRSUB record the affected entries and reverse their updates during backtracking. MONOSEMMCE saves and restores the previous double values, avoiding dependence on the order of floating-point subtraction; it also reuses depth-indexed buffers for candidate intersections and member contributions.

Parallel Ownership. OpenMP scheduling follows disjoint units already established by the algorithms. STRUCTBK, SEMBK, and STRSUB dynamically assign degeneracy roots to workers. For a large fixed-size subset layer, STRSUB further partitions combinations by their smallest selected vertex; workers read the frozen larger-size local maxima and merge their same-size results after the layer. MONOSEMMCE dynamically assigns feasible edge roots, each of which owns one canonical tree. Accumulators, recursion buffers, counters, and candidate output remain thread-local during search.

Result Consolidation. Both proposed methods use the grouped containment filter from Algorithm 2. Candidates are sorted once by decreasing size and deduplicated lexicographically. Queries within one equal-size group read a frozen inverted index and run in parallel; retained candidates are inserted only after the group completes. Consequently, synchronization occurs at result merging and between size groups rather than on recursive search states.

6 EXPERIMENTS

We organize the evaluation around seven research questions. RQ1 evaluates end-to-end performance and output agreement. RQ2 studies threshold sensitivity, RQ3 graph-size scalability, RQ4 parallel scalability, and RQ5 the search and engineering mechanisms isolated by ablation. RQ6 characterizes the MSC output space, while RQ7 compares the MSC definition with pairwise semantic constraints and relaxed structural pattern families.

6.1 Experimental Setup

Methods and comparison scope. STRUCTBK implements BK with Tomita pivoting and degeneracy-ordered roots [4, 11, 12, 32]; it supplies a threshold-independent structural-time reference. SEMBK is the direct baseline: it uses the same MSC definition, threshold, similarity values, and strict-containment filter as STRSUB and MONOSEMMCE. Consequently, runtime speedups are claimed only among these three exact MSC methods. PAIRSIM-MCE keeps edges whose endpoint similarity reaches τ and then runs structural MCE; it represents the every-pair similarity contract used by related constrained enumeration models [37]. FastQC enumerates maximal quasi-cliques under a degree-density constraint [38], while FaPlex enumerates maximal 2-plexes that may omit one incident edge per vertex [40]. These methods define different pattern families and are used only for output-family comparisons. Thus the runtime comparison is restricted to the three exact MSC methods, while the external methods provide output-contract reference points. **Datasets.** The common grid contains three graphs with natural attributes and five public graphs with reproducibly generated attributes. FB15K-237 descriptions use all-MiniLM-L6-v2 embeddings; GitHub-Social uses its official 4,005-dimensional developer features; and ogbn-arxiv

Dataset	$ V $	$ E $	D	Attributes
FB15K-237	14,541	248,611	384	natural
GitHub-Social	37,700	289,003	4,005	natural
ogbn-arxiv	169,343	1,157,799	128	natural
email-Enron	36,692	183,831	256	generated
sc-nasasrb	54,870	1,311,227	256	generated
sc-pkustk11	87,804	2,565,054	256	generated
sc-pwtk	217,891	5,653,221	256	generated
sc-lldoor	909,537	20,770,807	256	generated

Table 1: Public graphs and vector attributes used in the evaluation.

retains its official 128-dimensional node features [17, 33]. For email-Enron and the four SuiteSparse graphs [9, 22], the PCG64 pseudorandom generator draws 256 coordinates independently from $U[-1, 1)$ under fixed dataset seeds, followed by ℓ_2 normalization. Every graph topology is extracted directly from its public source dataset. Generated vectors provide controlled inputs for runtime, scale, and mechanism experiments; the semantic-output analyses in RQ7 use only the two graphs with official natural attributes.

Protocol. Experiments run on an AMD Ryzen 9 7945HX workstation with 16 physical cores, 32 hardware threads, and 32 GiB memory. We compile with GCC 13.2 and OpenMP. Each process is limited to 16 GiB and 32 OpenMP threads. The uniform timeout is 1,200 seconds. Completed runtime configurations are repeated three times and reported by the median; time-limit exceeded (TLE) denotes one complete 1,200-second limit window. Runtime binaries contain no statistics counters.

Absolute cosine values are not comparable across vector sources. We therefore derive thresholds from the edge-similarity distribution of each dataset. The common grid fixes q_{20} , q_{80} , and q_{99} before execution; sensitivity experiments use q_5 , q_{20} , q_{50} , q_{70} , q_{80} , q_{90} , q_{95} , q_{97} , and q_{99} . All exact methods receive the same stored float32 edge similarities and evaluate inclusive feasibility using double-precision sums.

6.2 RQ1: Exactness and End-to-End Efficiency

This experiment answers RQ1 by comparing complete MSC output sets and total runtime under the common grid. Exhaustive oracles first test 225,000 vertex-bound instances and 40,001 canonical-parent instances, including negative similarities, equality at τ , and tied minimum contributions. On real graphs, jointly completed methods agree on normalized clique sets, counts, and size histograms. In particular, all three exact methods return 263,721, 71,118, and 2,636 MSCs on GitHub-Social at q_{20} , q_{80} , and q_{99} , respectively; STRSUB and MONOSEMMCE also agree on every jointly completed row of Table 2. Across the eight graphs, the structural reference enumerates between 22,612 and 1,041,789 structural maximal cliques; these counts provide the front-end search scale behind the runtime comparisons.

Table 2 shows that STRSUB completes 22 of 24 workloads and every q_{20} row, while MONOSEMMCE resolves the two intermediate rows where host-clique subset enumeration reaches TLE. At q_{99} , canonical search is faster on all eight graphs, with speedups from 2.92 $\times$ on sc-lldoor to 6.24 $\times$ on email-Enron. SEMBK completes the six GitHub-Social and ogbn-arxiv rows but reaches TLE on all

Dataset	q	MSCs	STRUCTBK structure	SEMBK	STRSUB	MONOSEMCE
FB15K-237	20	206,470	0.103	TLE	0.221	TLE
	80	76,106		TLE	9.132	TLE
	99	7,476		TLE	45.234	10.091
GitHub-Social	20	263,721	0.110	55.269	0.327	29.315
	80	71,118		44.161	0.314	20.551
	99	2,636		38.213	0.391	0.118
ogbn-arxiv	20	925,239	0.309	2.783	1.233	3.357
	80	226,936		1.206	0.872	1.610
	99	10,261		1.318	0.843	0.197
email-Enron	20	223,703	0.090	18.330	0.259	16.739
	80	60,352		21.369	1.266	0.083
	99	1,819		9.608	0.206	0.033
sc-nasasrb	20	28,154	0.090	TLE	0.301	TLE
	80	834,914		TLE	102.820	0.759
	99	12,742		TLE	0.415	0.079
sc-pkustk11	20	22,612	0.143	TLE	0.645	TLE
	80	2,214,411		TLE	TLE	2.219
	99	24,891		TLE	0.611	0.167
sc-pwtk	20	58,523	0.286	TLE	1.093	TLE
	80	4,483,887		TLE	TLE	3.977
	99	54,907		TLE	1.364	0.421
sc-ldoor	20	274,585	1.679	TLE	4.870	TLE
	80	13,473,317		TLE	563.996	14.944
	99	202,030		TLE	4.845	1.658

Table 2: Median end-to-end runtime on the 24-workload common grid.

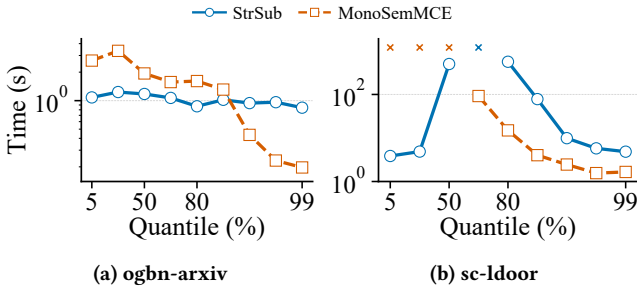


Figure 5: Threshold sweeps reveal complementary runtime regimes.

three FB15K-237 rows; its pivot-free search is slower than the best specialized method on every completed row. The gray STRUCTBK column is reported only to separate structural enumeration cost from semantic processing.

6.3 RQ2: Threshold Sensitivity

This experiment varies semantic selectivity on fixed graphs. Figure 5 uses the full fixed quantile grid. On ogbn-arxiv, both specialized methods complete the sweep and cross near q_{95} ; on sc-ldoor, STRSUB covers low and intermediate thresholds, while MONOSEMCE becomes dominant once the feasible canonical tree contracts. TLE points are shown at the fixed limit and are not interpolated.

6.4 RQ3: Scalability with Graph Size

Figure 6 repeats the comparison on nested induced ogbn-arxiv graphs whose vertices follow one fixed degree order. The crossover remains at q_{95} from 25K vertices through the full graph. At q_{99} , MONOSEMCE is 6.0 $\times$, 5.6 $\times$, 4.4 $\times$, and 3.5 $\times$ faster than STRSUB at

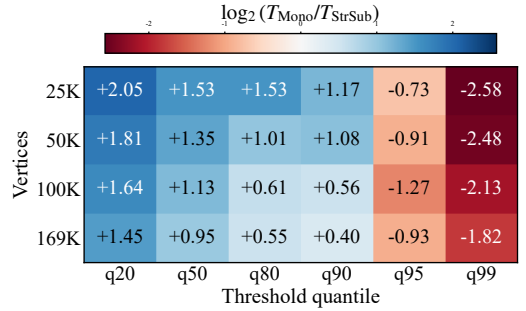


Figure 6: The selective-threshold advantage persists across graph sizes.

25K, 50K, 100K, and 169K vertices, respectively. The stable boundary links the regime change to semantic selectivity rather than one graph size.

6.5 RQ4: Parallel Scalability

Figure 7 reports one representative workload for each specialized algorithm. Dynamic edge-root scheduling gives MONOSEMCE 5.62 $\times$ speedup at 32 threads. Fixed-size subset partitioning gives STRSUB 2.39 $\times$ at 16 threads and 2.26 $\times$ at 32 threads. Separately collected phase counters attribute 98.88% of its 32-thread work to host-clique subset processing, while the final global filter takes 65 ms; the remaining saturation is therefore caused by unequal host-clique tasks.

6.6 RQ5: Search and Engineering Mechanisms

Figure 8 relates runtime regimes to work counters collected outside timing runs. On sc-nasasrb at q_{99} , the sorted-edge bound reduces

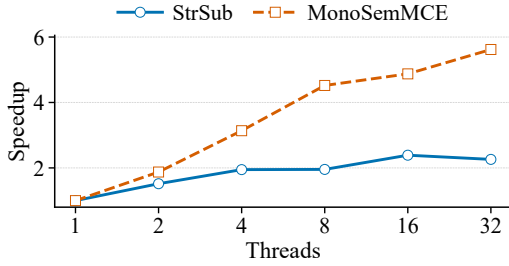


Figure 7: Parallel speedup from independent search-root scheduling.

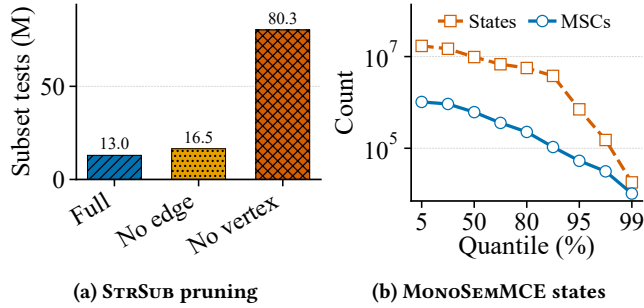


Figure 8: Bounds and canonical ownership reduce explored states.

Component	Metric / workload	Full	Removed
Sorted-edge bound	tests (M) / nasa q_{99}	12.95	16.50
Vertex bound	tests (M) / nasa q_{99}	12.95	80.28
Local containment	GiB / FB15K q_{80}	0.12	2.26

Table 3: Effect of the three STRSUB pruning mechanisms.

subset tests from 16.50 to 12.95 million, and the vertex bound reduces them from 80.28 to 12.95 million. On ogbn-arxiv, accepted canonical states contract from 17.08 million at q_5 to 18,011 at q_{99} . The ablations proceed hierarchically: they first isolate semantic pruning in STRSUB, then duplicate-state organization in MONOSEMCE, and finally the shared graph-engine primitives while holding the exact output fixed.

The corresponding runtimes are 0.287 versus 0.324 seconds for the sorted-edge bound, 0.287 versus 0.312 seconds for the vertex bound, and 8.060 versus 11.685 seconds for local containment. Table 4 then holds the feasibility test, similarity cache, and final filter fixed while replacing only duplicate-state organization. *Visited DFS* suppresses duplicates with a global state table, whereas *Layered* materializes and deduplicates all feasible cliques of one size before generating the next. Canonical ownership is fastest and uses the least memory on both workloads; every variant returns the same normalized MSC set and size histogram.

Table 5 isolates the shared graph engine and the incremental semantic state. We report the median search time over three runs; all variants return the same output count, and the exact normalized output sets also match on the correctness workload. Reordering has

Generation	ogbn q_{95}		nasa q_{80}	
	s	GiB	s	GiB
Canonical	0.388	0.39	0.729	0.22
Visited DFS	2.432	0.53	3.069	0.64
Layered	1.635	0.49	3.922	0.81

Table 4: Canonical ownership versus duplicate-state organizations.

Method	Workload	Removed	Full (s)	Removed (s)	Ratio
STRSUB	ldoor q_{99}	Reorder	4.229	4.330	1.02
STRSUB	ldoor q_{99}	Bitmap	4.229	4.701	1.11
MONOSEMCE	ldoor q_{99}	Reorder	1.111	1.129	1.02
MONOSEMCE	ldoor q_{99}	Bitmap	1.111	1.026	0.92
MONOSEMCE	GitHub q_{80}	Accumulator	23.256	32.752	1.41

Table 5: Engineering ablation; ratio is removed over full runtime.

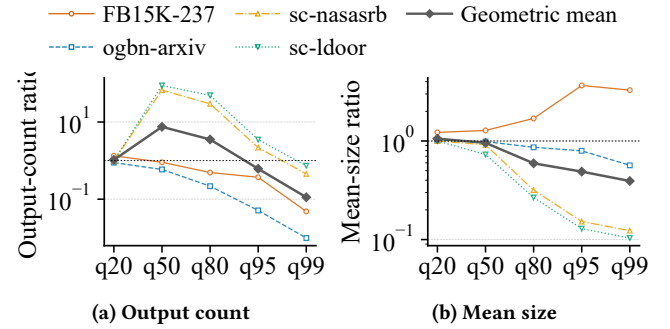


Figure 9: MSC output expands and then contracts as selectivity increases.

a small positive effect for both search organizations. Sparse blocked bitmaps accelerate the repeated BK phase of STRSUB, but their block-merge overhead exceeds the benefit for the lighter neighbor operations in MONOSEMCE on this workload. Incremental accumulation has the largest engineering effect: recomputing each candidate’s contribution increases MONOSEMCE time by 1.41 $\times$. Thus the graph engine shares storage and kernels, while the useful representation depends on the access pattern of the search algorithm.

6.7 RQ6: MSC Output Space

This experiment measures exact output counts and mean sizes on four graphs: FB15K-237, ogbn-arxiv, sc-nasasrb, and sc-ldoor. It uses five quantiles from q_{20} through q_{99} . Figure 9 normalizes each value by STRUCTBK on the same graph. The geometric-mean count ratios are 1.04, 7.50, 3.53, 0.61, and 0.11: intermediate thresholds split large structural cliques into many feasible subsets that are maximal by inclusion. Selective thresholds instead contract the result family. Mean size decreases from 1.06 to 0.39 of the structural reference over the same grid.

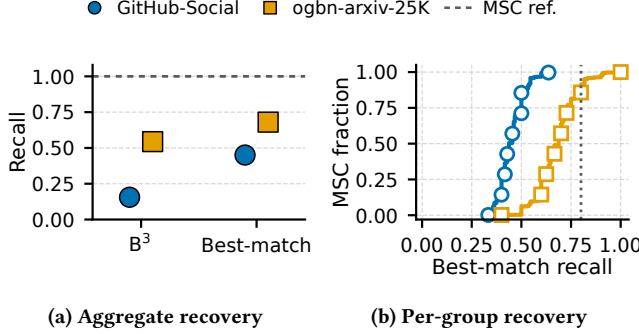


Figure 10: Pairwise thresholding fragments collectively feasible MSCs.

6.8 RQ7: Group Preservation and External Pattern Families

RQ7 first compares MSC with the every-pair definition. The reference cover contains all size-at-least-ten MSCs at q_{80} on GitHub-Social and ogbn-arxiv-25K; the response cover contains every non-singleton PAIRSIM-MCE output at the same τ . For these overlapping covers, Extended B^3 Recall is the overlap-aware, entity-level recall of the reference cover and measures collective fragmentation [1]. Best-match recall $R_{\max}(C) = \max_{P \in \mathcal{P}} |C \cap P|/|C|$ measures the largest fraction of MSC C retained by one output [16, 35].

Figure 10 shows the definition effect. PAIRSIM-MCE obtains B^3 Recall/best-match recall of 0.156/0.450 on GitHub-Social and 0.544/0.680 on ogbn-arxiv-25K. None of 605 GitHub-Social MSCs and only 0.32% of 624 ogbn-arxiv-25K MSCs is retained intact; respectively, 100% and 84.9% lose at least 20% of their members. The every-pair constraint therefore fragments groups that remain complete and collectively feasible.

The external comparison fixes FastQC at $\gamma = 0.9$, FaPlex at $k = 2$, and uses STRUCTBKas the structural reference on the same q_{80} graphs. For two groups A and B , their match score is $F_1(A, B) = 2|A \cap B|/(|A| + |B|)$ [35]. *Recovery* averages each MSC’s best match in the baseline family, whereas *fidelity* averages each baseline result’s best match in the MSC family. This directional pair measures output-family agreement rather than ground-truth clustering.

Figure 11 reports high recovery but low fidelity; MSC is the reference family and its trivial self-match is therefore omitted. The ranges are 0.851–0.992 and 0.123–0.269, respectively. High recovery is expected because every MSC extends to a maximal clique, quasi-clique, or 2-plex; fidelity exposes the additional patterns introduced by that extension. FastQC and FaPlex are not structural MCE duplicates: their pair-weighted densities are 0.942–0.952, fewer than 0.05% of retained outputs are cliques, and they cover $12.5 \times$ – $27.6 \times$ more vertices with lower cosine.

A size-18 ogbn-arxiv MSC consists entirely of cs.IT papers on private information retrieval. Its average is 0.919 at $\tau = 0.889$, although 20 of 153 pairs fall below τ , illustrating the collective group preserved here.

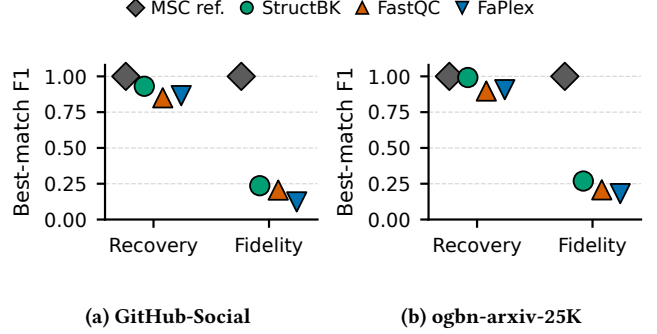


Figure 11: Relaxed families recover MSCs but add many unmatched results.

7 RELATED WORK

Clique Enumeration and Listing. BK backtracking [4], Tomita pivoting [32], and degeneracy ordering [11, 12] underpin exact MCE. The classical output bound and reporting perspective were established by Moon–Moser and subsequent maximal-clique analyses [5, 26]. Recent methods reduce BK subproblems [10, 34] or summarize clique families [24]; these methods preserve structural maximality and do not impose a group-level vector constraint. Large-graph implementations also exploit specialized search structures and parallel execution [6, 8, 23].

Constrained Cohesive Subgraphs. FastQC [38] and FaPlex [40] enumerate maximal quasi-cliques and k -plexes, respectively, while biplex methods relax complete bipartite structure [7]. They change the adjacency contract, whereas MSC retains a complete induced subgraph.

The closest constrained-enumeration model is maximal similar-biclique enumeration [37]: its hereditary constraint requires every vertex pair on one side to satisfy a similarity threshold. Similar-weight bicliques likewise constrain edge weights in a bipartite pattern [36]. MSC differs along both axes: it operates on cliques in a general graph and constrains the average of all pairwise vector similarities. This aggregate condition admits low-similarity pairs and is non-monotone under vertex insertion and deletion, which is precisely the source of the search problem studied here.

Attributed Community Search. Attributed-graph mining often couples a relaxed dense subgraph with an attribute set or correlation score [30], while CODICIL returns a partition of a content-augmented graph [29] and attributed truss community search returns a query-specific group [18]. Feature-subspace [15, 27] and cross-graph community methods [13] also optimize a partition or query-oriented objective. Structural/attribute clustering and attributed community search provide related but query- or partition-oriented alternatives [14, 39]. These outputs are not inclusion-wise maximal complete groups under a vector aggregate; MSC therefore specifies a distinct enumeration contract rather than renaming an attributed community.

Vector Attributes. Representation models [3, 21, 28, 31] can supply the vectors. Given these fixed attributes, MSC evaluates the

aggregate threshold exactly; approximate vector retrieval addresses a separate nearest-neighbor task [2, 19, 25].

8 CONCLUSION

We formulated exact MSC enumeration under an average pairwise-similarity constraint and showed that its associated size decision problem is NP-complete. STRSUB combines structural decomposition with admissible semantic bounds; MONOSEMCE turns feasible minimum-contribution deletion into a canonical reverse-search forest. Their complementary threshold regimes jointly complete all 24 common-grid workloads, with MONOSEMCE $2.92\times\text{--}6.24\times$ faster than STRSUB on every q_{99} row. Ablations connect these gains to layer pruning and canonical-state contraction, while natural-attribute results show when averaging preserves complete groups fragmented by pairwise thresholding. These results connect the group-level semantic definition to complementary exact search organizations.

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